Program : **Diploma in Chemical Engineering**	
Course Code : **4079**	Course Title: **Mechanical Operations Lab**
Semester : **4**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L: 0 T: 0 P: 3)**	Periods per semester: **45**

Course Objectives:

- To provide hands on experience in analyzing the size reduction and separation of particles.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Size separation, size reduction and sedimentation		Mechanical operations	3

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate sieve analysis.	12	Applying
CO2	Illustrate sub sieve analysis.	9	Applying
CO3	Illustrate sedimentation and filtration.	15	Applying
CO4	Illustrate size reduction.	9	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3			3	3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	**Illustrate sieve analysis.**		
M1.01	Estimate the average diameter of solid particles using sieve analysis.	3	Applying
M1.02	Estimate the effectiveness of sieve.	3	Applying
M1.03	Estimate the effectiveness of vibrating screen.	3	Applying
M1.04	Open ended experiments	3	Applying
CO2	**Illustrate sub sieve analysis**		
M2.01	Estimate the particle size distribution of particulate solid using beaker decantation method.	3	Applying
M2.02	Estimate the particle size distribution of particulate solid using pipette analysis.	3	Applying
M2.03	Open ended experiments	3	Applying
	Series Test – I		
CO3	**Illustrate sedimentation and filtration.**		
M3.01	Estimate the area of continuous thickener by batch sedimentation experiment.	3	Applying
M3.02	Estimate the specific cake resistance of given slurry using leaf filter.	3	Applying
M3.03	Estimate the specific cake resistance of given slurry using filter press.	3	Applying
M3.04	Estimate the separation efficiency of a given cone classifier.	3	Applying
M3.05	Open ended experiments	3	Applying

CO4	**Illustrate size reduction.**		
M4.01	Verify the law of crushing using ball mill	3	Applying
M4.02	Open Ended Experiments	6	Applying
	Series Test – II		

Text / Reference

T/R	Book Title/Author
T1	Unit Operations of Chemical Engineering, Vol. 1 & 2; Chattopadhyay O.P
R1	Introduction to Chemical Engineering; Walter.L.Badger & Julius.T.Banchero
R2	Unit Operations – I ; K A Gavhane

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103107123/